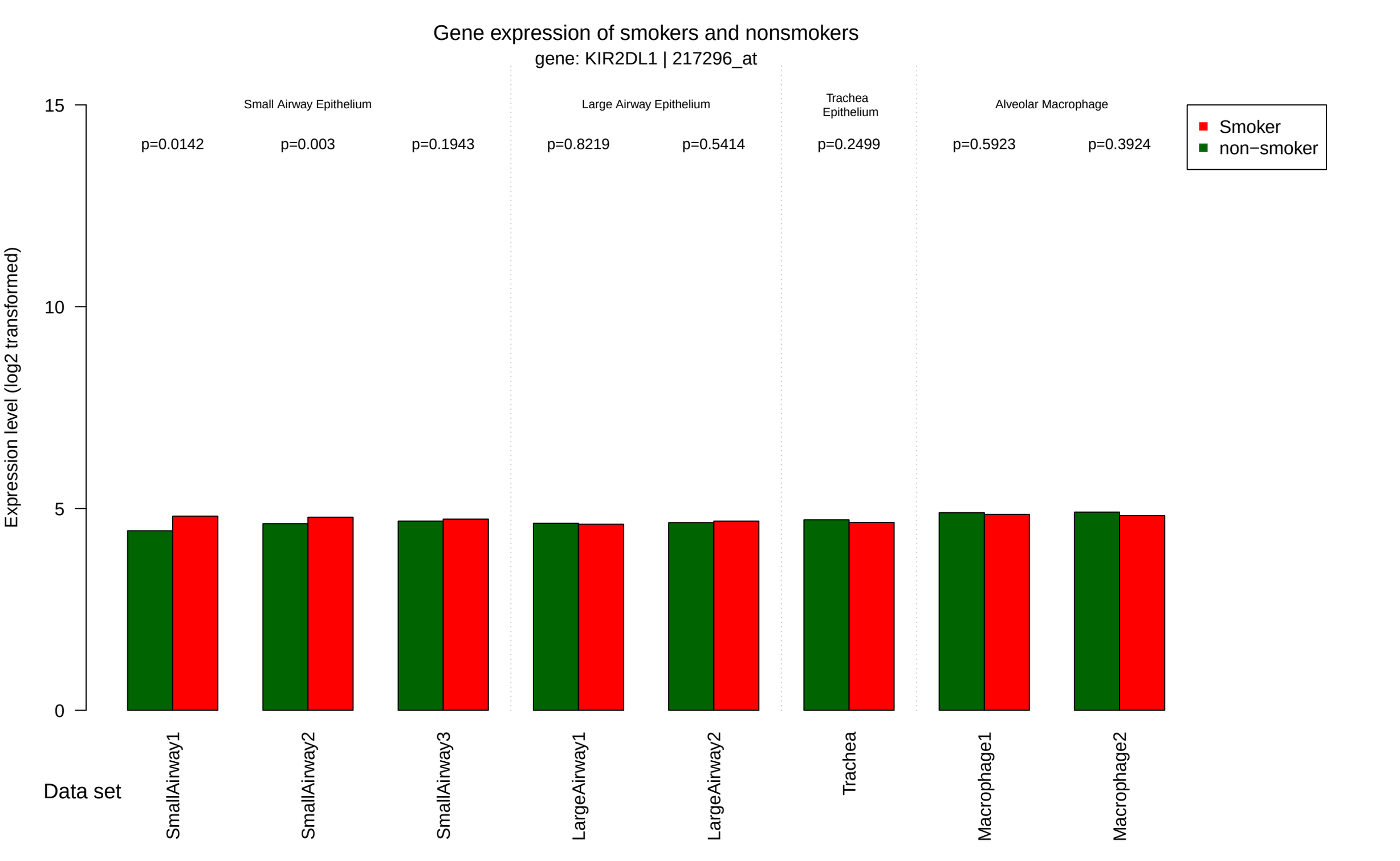
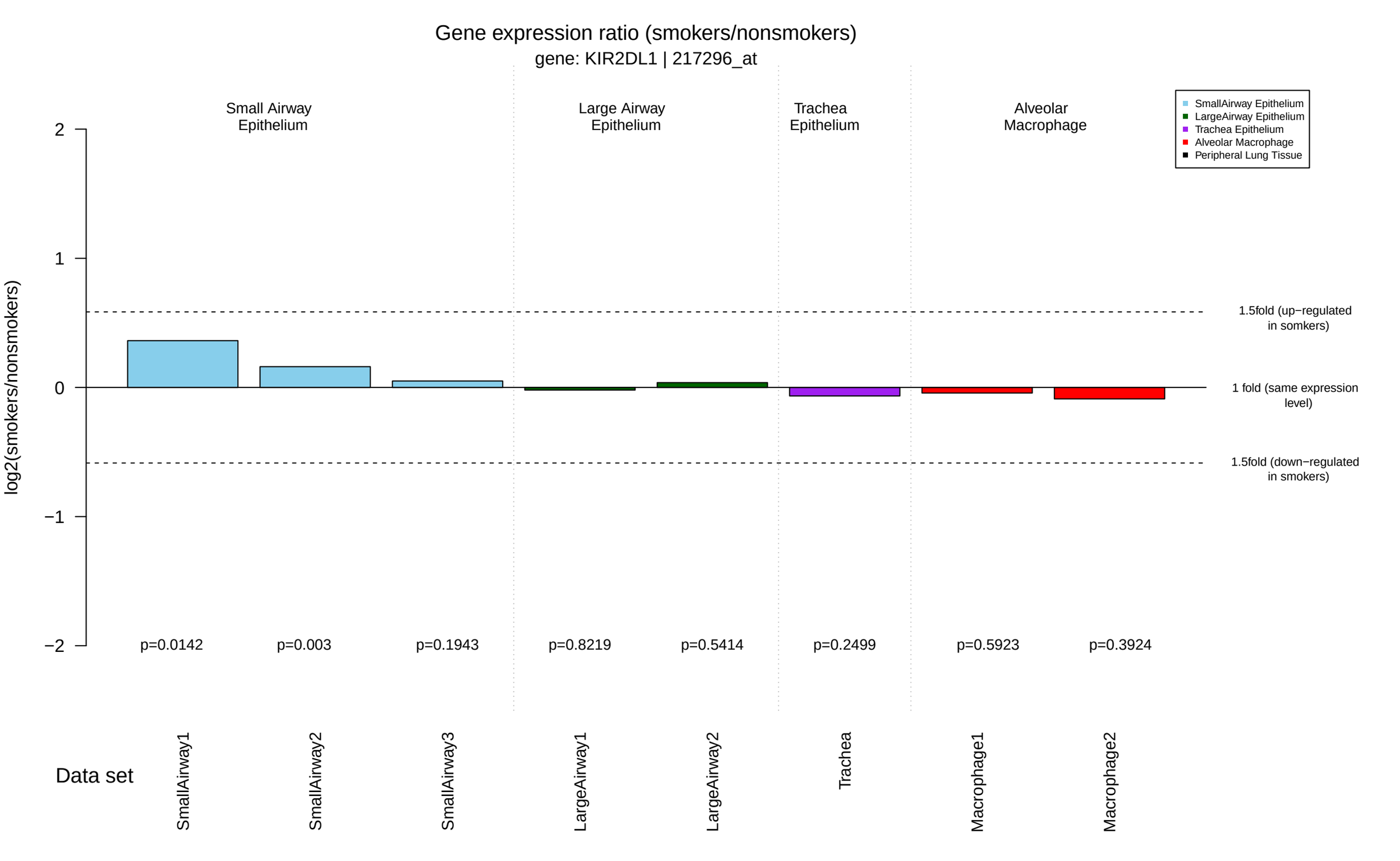
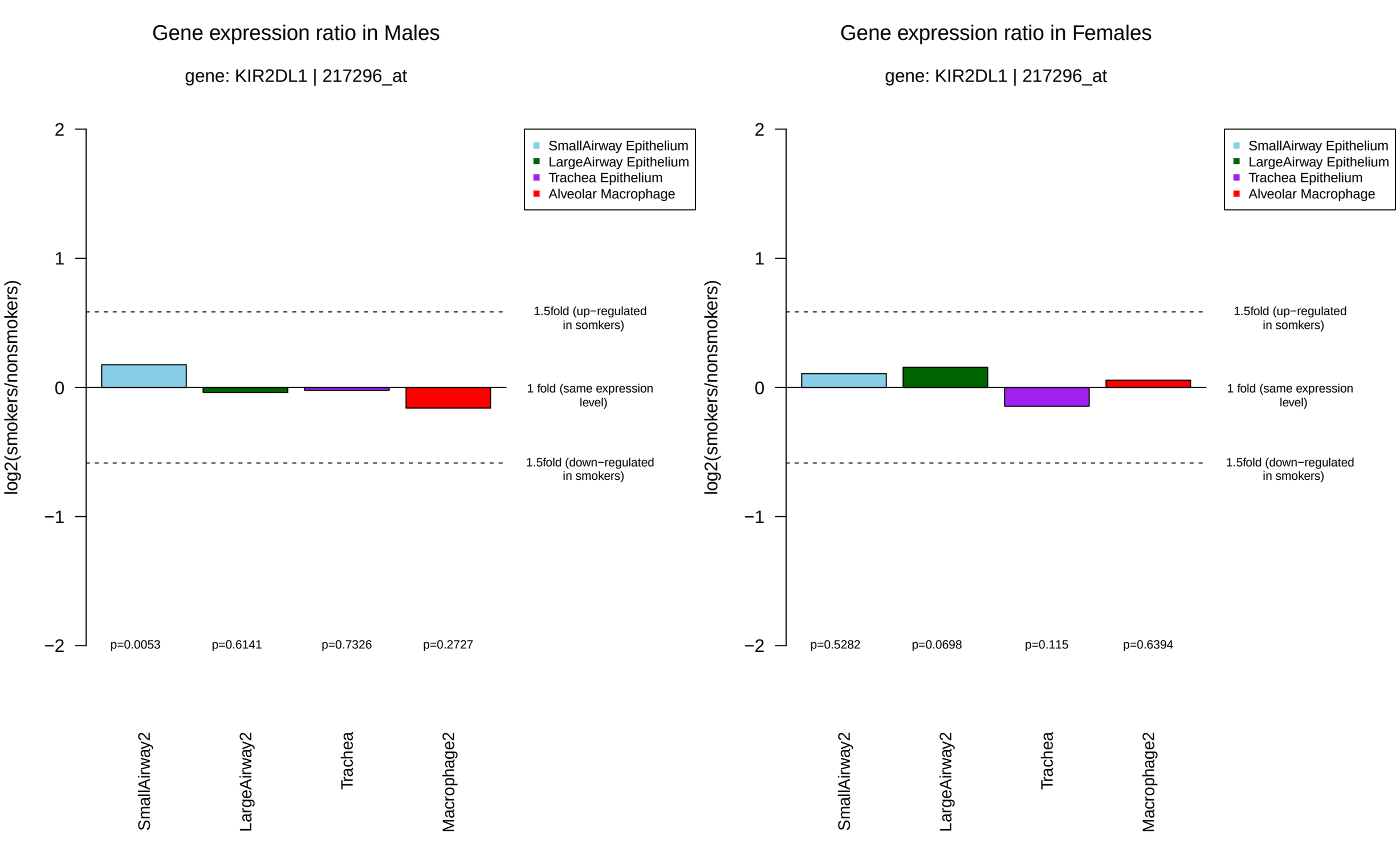
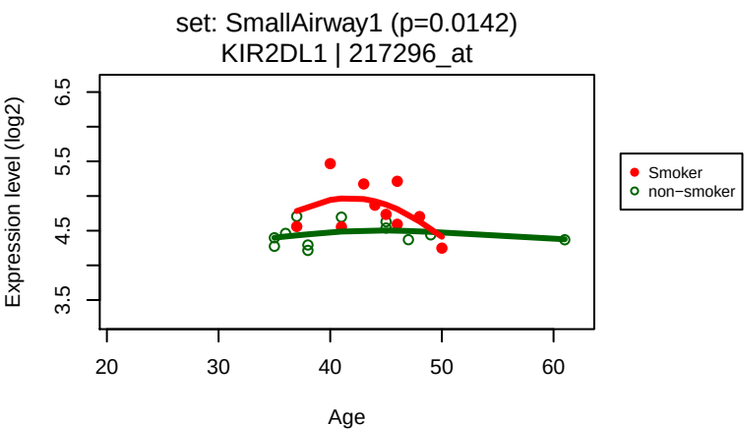
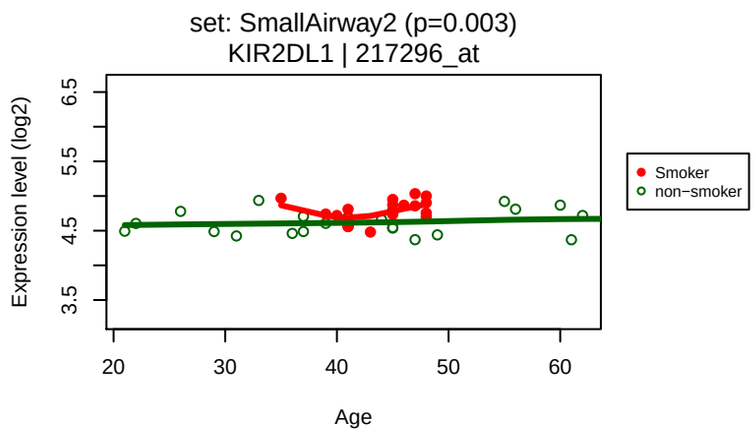
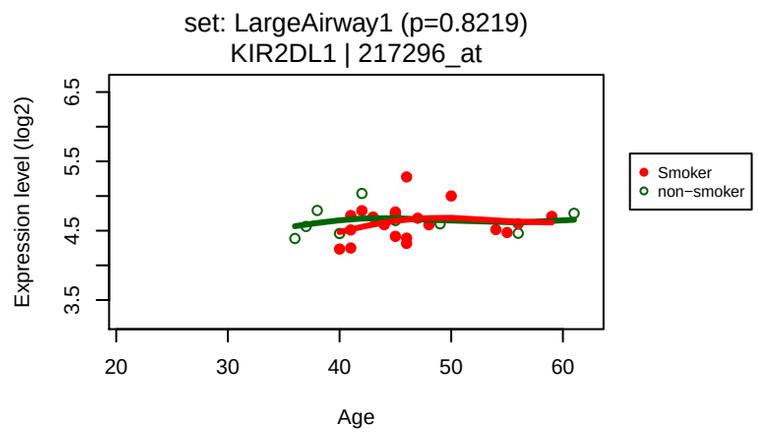

Gene expression of smokers and nonsmokers
gene: KIR2DL1 | 217296_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0142
p=0.003
p=0.1943
p=0.8219
p=0.5414
p=0.2499
p=0.5923
p=0.3924
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

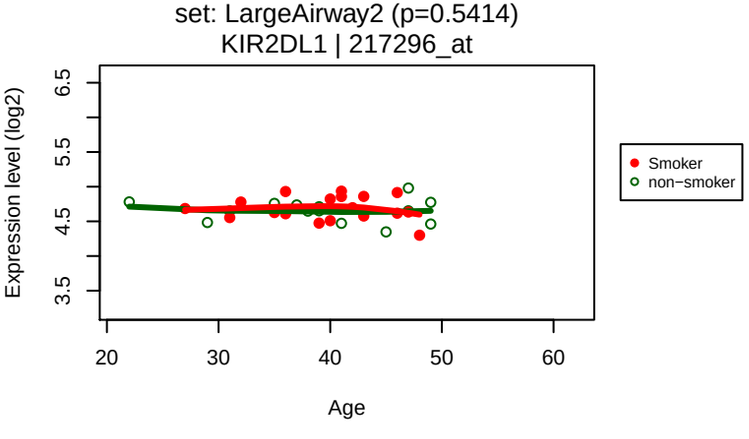

Gene expression ratio (smokers/nonsmokers)
gene: KIR2DL1 | 217296_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0142
p=0.003
p=0.1943
p=0.8219
p=0.5414
p=0.2499
p=0.5923
p=0.3924
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

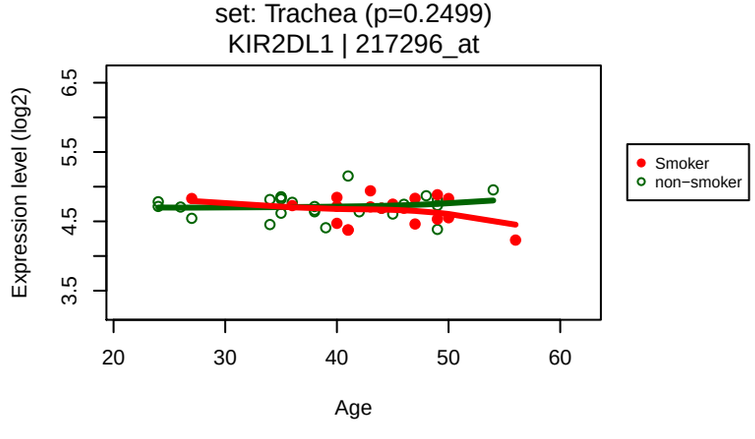

Gene expression ratio in Males
gene: KIR2DL1 | 217296_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0053
p=0.6141
p=0.7326
p=0.2727
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: KIR2DL1 | 217296_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5282
p=0.0698
p=0.115
p=0.6394
SmallAirway2
LargeAirway2
Trachea
Macrophage2

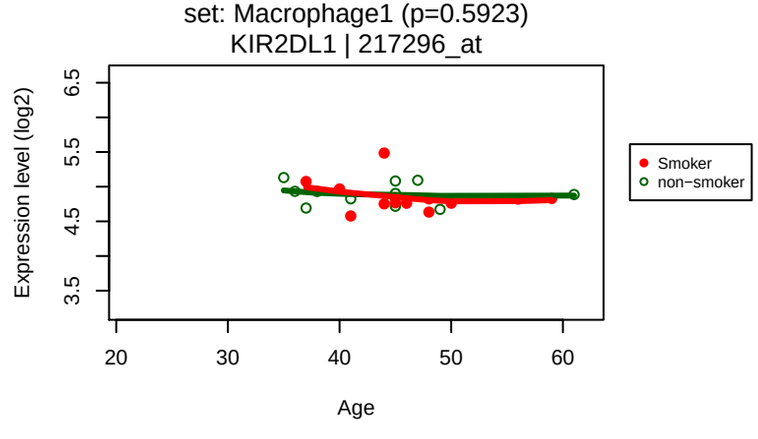

set: SmallAirway1 (p=0.0142)
KIR2DL1 | 217296_at
Expression level (log2)
3.5
4.5
5.5
6.5
20
30
40
50
60
Age
Smoker
non−smoker

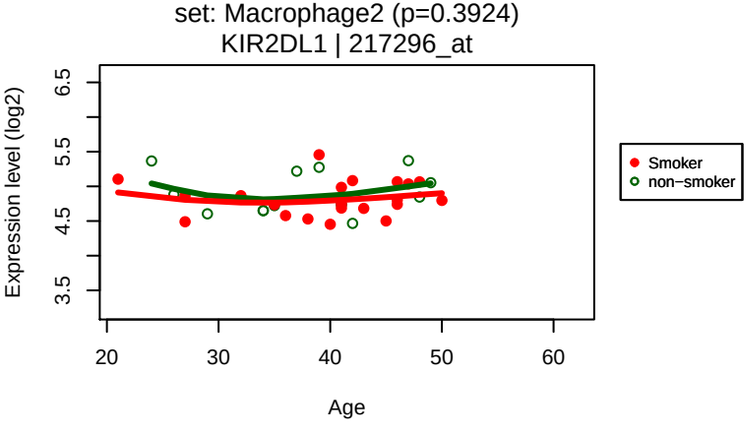

set: SmallAirway2 (p=0.003)
KIR2DL1 | 217296_at
Expression level (log2)
3.5
4.5
5.5
6.5
20
30
40
50
60
Age
Smoker
non−smoker

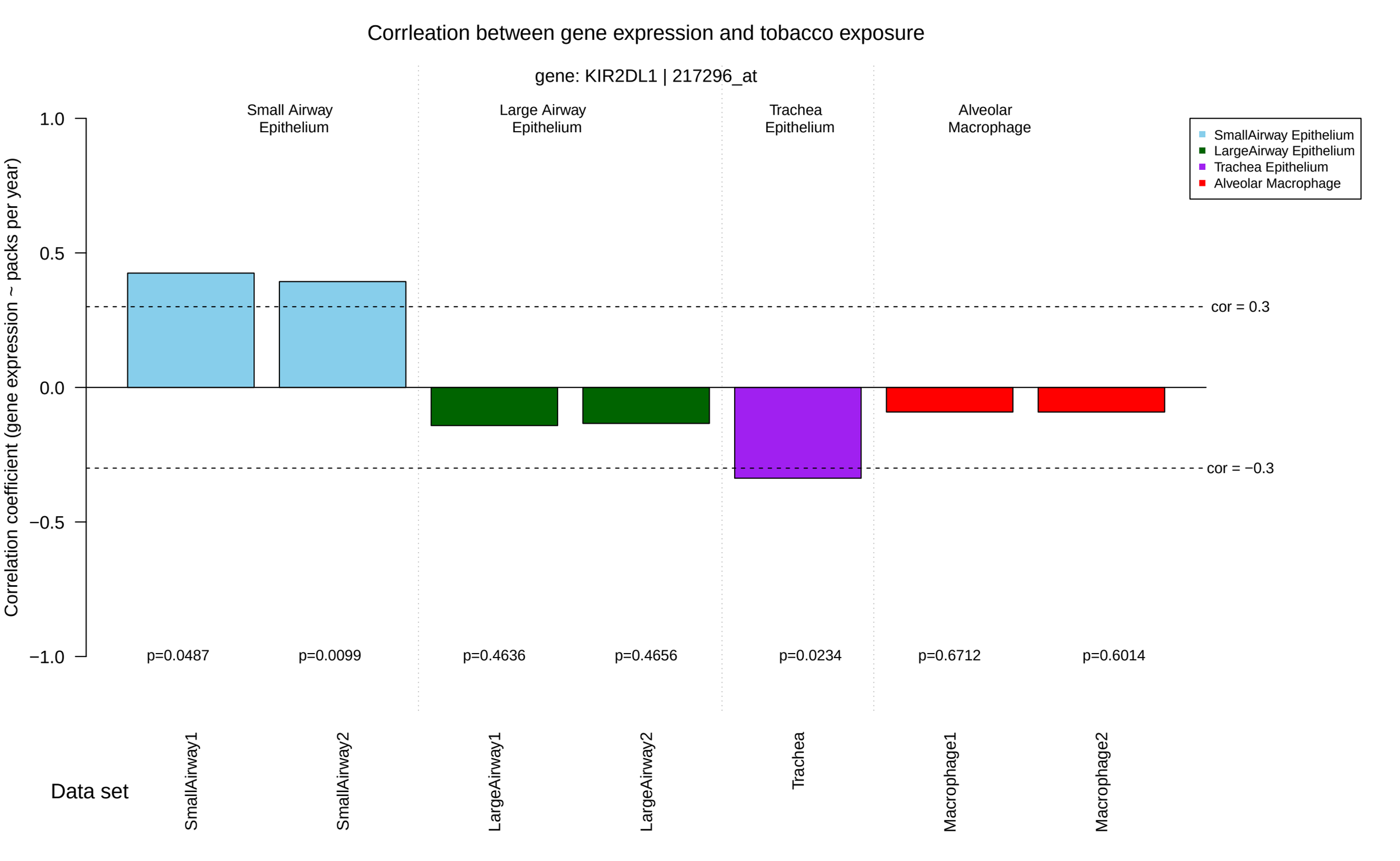

set: LargeAirway1 (p=0.8219)
KIR2DL1 | 217296_at
Expression level (log2)
3.5
4.5
5.5
6.5
20
30
40
50
60
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.5414)
KIR2DL1 | 217296_at
Expression level (log2)
3.5
4.5
5.5
6.5
20
30
40
50
60
Age
Smoker
non−smoker


set: Trachea (p=0.2499)
KIR2DL1 | 217296_at
Expression level (log2)
3.5
4.5
5.5
6.5
20
30
40
50
60
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5923)
KIR2DL1 | 217296_at
Expression level (log2)
3.5
4.5
5.5
6.5
20
30
40
50
60
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3924)
KIR2DL1 | 217296_at
Expression level (log2)
3.5
4.5
5.5
6.5
20
30
40
50
60
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: KIR2DL1 | 217296_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0487
p=0.0099
p=0.4636
p=0.4656
p=0.0234
p=0.6712
p=0.6014
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2